

detect-white: Chen & Storey matrix, diagonal free, then whitened

genotypes G

Chen & Storey H

L : diag. imputed by
top 3 PCs (rank 4)

S : the diagonal
(small: h_i already removed)

one pass

fit
 $H = L + S$

noise from the raw Gram

Σ : $v = \text{raw diag.} - L$
($\approx$ heterozygosity)

whitened:
 $A_{ij}/\sqrt{v_i v_j}$

PCs: PC2+1 vs PC3+1
MXL axis $R^2 = 1.00$

$\Sigma^{-1/2} A \Sigma^{-1/2}$

eigen-
vectors,
 $\times \Sigma^{1/2}$

PC3+1

PC2+1

+

